Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The value of $1^2 \cdot C_1 + 3^2 \cdot C_3 + 5^2 \cdot C_5 + \ldots$ is :

(A) $n(n-1)^{n-2} + n \cdot 2^{n-1}$

(B) $n(n-1)^{n-2}$

(C) $n(n-1)^{n-3}$

(D) None of these

Q.2

In how many ways vertices of a square can be coloured with 4 distinct colour if rotations are considered to be equivalent, but reflections are distinct?

(A) 65

(B) 70

(C) 71

(D) None of these

Q.3

The total number of 5 digit numbers of different digits in which the digit in the middle is the largest, is

(A) $\sum_{n=4}^{9} {}^{n}P_4$

(B) $\sum_{n=4}^{9} {}^{n}P_4 - \frac{1}{3!}\sum_{n=3}^{9} {}^{n}P_3$

(C) $30(3!)$

(D) None of these

Q.4

If $n = 2^{p-1}\left(2^p - 1\right)$, where $2^p - 1$ is a prime, then the sum of the divisors of n is equal to

(A) n

(B) $2n$

(C) pn

(D) p^{n}

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Q.5

Let p be a prime number such that $\mathrm{p} \geq 11$. Let $n = p! + 1$. The number of primes in the list $n + 1, n + 2, n + 3, \ldots, n + p - 1$, is

(A) $\mathrm{p} - 1$

(B) 2

(C) 1

(D) None of these

Q.6

If $S = (1)(1!) + (2)(2!) + (3)(3!) + \ldots. + n(n!)$, then

(A) $\frac{S+1}{n!} \in$ integer

(B) $\frac{S+1}{n!} \notin$ integer

(C) $\frac{S+1}{n!}$ cannot be discussed

(D) None of these

Q.7

The number of ways in which a mixed doubles game in tennis can be arranged from 5 married couples, if no husband and wife play in the same game, is

(A) 46

(B) 54

(C) 60

(D) None of these

Q.8

The expression $^{\mathrm{n}}\mathrm{C}_{\mathrm{r}} + 4.^{\mathrm{n}}\mathrm{C}_{\mathrm{r}-1} + 6.^{\mathrm{n}}\mathrm{C}_{\mathrm{r}-2} + 4.^{\mathrm{n}}\mathrm{C}_{\mathrm{r}-3} + ^{\mathrm{n}}\mathrm{C}_{\mathrm{r}-4}$ is equal to

(A) $^{\mathrm{n}+4}\mathrm{C}_{\mathrm{r}}$

(B) $2.^{\mathrm{n}+4}\mathrm{C}_{\mathrm{r}-1}$

(C) $4.^{\mathrm{n}}\mathrm{C}_{\mathrm{r}}$

(D) $11.^{\mathrm{n}}\mathrm{C}_{\mathrm{r}}$

Q.9

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Total number of required functions is $f:\{1,2,3,4,5\}\to\{1,2,3,4,5\}$ that are onto and $f(i)\neq i$, is equal to

(A) 9

(B) 44

(C) 16

(D) None of these

Q.10

There are 4 letters and 4 directed envelopes. The number of ways in which all the letters can be put in wrong envelope is

(A) 9

(B) 4

(C) 5

(D) 12

Q.11

A person writes letters to six friends and addresses the corresponding envelopes. Let x be the number of ways so that at least two of the letters are in wrong envelopes and y be the number of ways so that all the letters are in wrong envelopes. Then $x-y=$

(A) 719

(B) 265

(C) 454

(D) None

Q.12

Ravish writes letters to his five friends and addresses the corresponding envelopes. In how many ways can the letters be placed in the envelopes so that at least two of them are in the wrong envelopes?

(A) 109

(B) 118

(C) 119

(D) None of these

Q.13

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The number of distinct rational numbers x such that $0 < x < 1$ and $x = \frac{p}{q}r$, where $\mathrm{p}, \mathrm{q} \in \{1, 2, 3, 4, 5, 6\}$, is

(A) 15

(B) 13

(C) 12

(D) 11

Q.14

The greatest common divisor of ${}^{20}\mathrm{C}_1, {}^{20}\mathrm{C}_3, \ldots\ldots, {}^{20}\mathrm{C}_{19}$ is

(A) 20

(B) 4

(C) 5

(D) None of these

Q.15

Let $E = (2n+1)(2n+3)(2n+5)\ldots..(4n-3)(4n-1)$; $n > 1$ then $2^n E$ is divisible by

(A) ${}^{n}C_{n/2}$

(B) ${}^{2n}C_{n}$

(C) ${}^{3n}C_{n}$

(D) ${}^{4n}C_{2n}$

Answers & Solutions

Q.1 Answer:

None of these

Solution:

We know, $\sum_{r=1}^{n} r^2 \cdot {}^{n}\mathrm{C}_r = n(n-1)2^{n-2} + n \cdot 2^{n-1}$
and $\sum_{r=1}^{n}(-1)^{r-1} \cdot r^2 \cdot {}^{n}\mathrm{C}_r = 0$
Adding $(1)\&(2)$ **we get.**
$2\left[1^2.\mathrm{C}_1 + 3^2 \cdot \mathrm{C}_3 + 5^2\mathrm{C}_5 + \ldots.\right] = n(n-1)2^{n-2} + n$
2^{n-1}
$\Rightarrow \left[1^2\mathrm{C}_1 + 3^2\mathrm{C}_3 + 5^2\mathrm{C}_5 + \ldots\right]$
$= n(n-1)2^{n-3} + n \cdot 2^{n-2}.$

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Q.2 Answer:

70

Solution:

Here in this case condition is similar to formation of necklace i.e.,
$(\mathrm{n}, \mathrm{k})=\frac{1}{n} \sum_{i=1}^{n} k^{\gcd(n,i)}$
We can use this formula or from the table (you shouldn't memorize it) required number of ways is 70

Q.3 Answer:

None of these

Solution:

Since the largest digit is in the middle, the middle digit is greater than or equal to 4, the number of numbers with 4 in the middle $= {}^{4}\mathrm{P}_{4}-{}^{3}\mathrm{P}_{3}$
($\because$ **the other four places are to be filled by** $0, 1, 2$ **and 3, and a number cannot begin with 0). Similarly, the numbers of numbers with 5 in the middle** $= {}^{5}\mathrm{P}_{4}-{}^{4}\mathrm{P}_{3}$**, etc.)** $\therefore$ **The required number of numbers** $= \left({}^{4}\mathrm{P}_{4}-{}^{3}\mathrm{P}_{3}\right)+\left({}^{5}\mathrm{P}_{4}-{}^{4}\mathrm{P}_{3}\right)+\left({}^{6}\mathrm{P}_{4}-{}^{5}\mathrm{P}_{3}\right)+$
$\ldots .+\left({}^{9}\mathrm{P}_{4}-{}^{8}\mathrm{P}_{3}\right)$
$=\sum_{n=4}^{9}{}^{\mathrm{n}}\mathrm{P}_{4}-\sum_{\mathrm{n}=3}^{8}{}^{\mathrm{n}}\mathrm{P}_{3}$

Q.4 Answer:

$2n$

Solution:

$S=\left(1+2^{p}-1\right)\left(2^{0}+2^{1}+\ldots .+2^{p-1}\right)$
$=2^{p} \frac{2^{p}-1}{2-1}$
$\Rightarrow \quad 2^{p}\left(2^{p}-1\right)=2 \times 2^{p-1}\left(2^{p}-1\right)=2N$

Q.5 Answer:

None of these

Solution:

For $1 \leq i \leq p-1, p!$ **is divisible by** $(1+1)$ **Thus,** $\mathrm{n}+\mathrm{i}=\mathrm{p}!+(\mathrm{i}+1)$ **is divisible by** $(\mathrm{i}+1)$ **for** $\leq \mathrm{i} \leq \mathrm{p}-1$
$\therefore$ **None of** $n+1, n+2, \ldots \ldots, n+p-1$ **is prime.**

Q.6 Answer:

$\frac{S+1}{n!} \in$ integer

Solution:

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We have,
$S = \sum_{k=1}^{n} k(k!) = \sum_{k=1}^{n} \{(k+1) - 1\}(k!)$
$= \sum_{k=1}^{n} \{(k+1)! - k!\} = (n+1)! - 1 \Rightarrow S + 1 = (n+1)$
Thus, $\frac{S+1}{n!} \in$ **integer.**

Q.7 Answer:

60

Solution:

Let the sides of the game be A **and** B**. Given**
5 married couples, i.e., 5 husbands and 5 wives. Now, 2 husbands for two sides A **and** B **can be selected out of** $5 = {}^{5}\mathrm{C}_{2} = 10$ **ways. After choosing the two husbands their wives are to be excluded (since no husband and wife play in the same game). So we have to choose 2 wives out of remaining** $5 - 2 = 3$ **wives i.e.** ${}^{3}\mathrm{C}_{2} = 3$ **ways.**
Again two wives can interchange their sides A **and** B **in** $2! = 2$ **ways. By the principle of multiplication, the required number of ways** $= 10 \times 3 \times 2 = 60$

Q.8 Answer:

${}^{\mathrm{n}+4}\mathrm{C}_{\mathrm{r}}$

Solution:

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$$
\begin{aligned}
&{}^{n}C_{r}+4.{}^{n}C_{r-1}+6.{}^{n}C_{r-2}+4.{}^{n}C_{r-3}+{}^{n}C_{r-4}\\
&=({}^{n}C_{r}+{}^{n}C_{r-1})+3\,({}^{n}C_{r-1}+{}^{n}C_{r-2})\\
&\quad+3\,({}^{n}C_{r-2}+{}^{n}C_{r-3})+({}^{n}C_{r-3}+{}^{n}C_{r-4})\\
&={}^{n+1}C_{r}+3.{}^{n+1}C_{r-1}+3.{}^{n+1}C_{r-2}+\\
&\quad{}^{n+1}C_{r-3}\\
&=({}^{n+1}C_{r}+{}^{n+1}C_{r-1})+2({}^{n+1}C_{r-1}+\\
&\quad{}^{n+1}C_{r-2})\\
&\qquad\qquad+({}^{n+1}C_{r-2}+{}^{n+1}C_{r-3})\\
&={}^{n+2}C_{r}+2\,.\,({}^{n+2}C_{r-1}+{}^{n+2}C_{r-2})\\
&=({}^{n+2}C_{r}+{}^{n+2}C_{r-1})+({}^{n+2}C_{r-1}+{}^{n+2}C_{r-2})\\
&={}^{n+3}C_{r}+{}^{n+3}C_{r-1}={}^{n+4}C_{r}
\end{aligned}
$$

Q.9 Answer:

44

Solution:

Total number of required functions = Number of dearrangement of 5 objects = $5!\left(\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}\right)=44$

Q.10 Answer:

9

Solution:

Let us first consider 2 letters and 2 envelopes, then there is only one way to place both the letters in wrong envelope. Next, we consider 3 letters and 3 directed envelopes. The number of ways of putting all letters in wrong envelopes

= Total number of possible arrangements − Number of ways in which all letters are in correct envelopes

- Number of ways in which 1 letter in correct envelope = $3!-1-{}^{3}C_{1}\times 1=2$

[∵ The case of two letters in correct envelope and one in wrong envelope is not possible] Further, we

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consider 4 letters and 4 directed envelopes. The number of ways of putting all letters in wrong envelopes = Total number of possible arrangements - number of ways in which all letters are in correct envelope Number of ways in which 1 letter is in correct envelopes (3 in wrong envelope) - Number of ways in which 2 letters are in correct envelope $(2$ in wrong envelope $) = 4! - 1 - {}^4C_1 \times 1 = 9$. NOTE : Such problems are called problems of derangement. Hence, using the formula of derangement. The required number of ways of placing all letters in wrong envelope $= 4!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{2!} + \frac{1}{4!}\right] = \frac{4!}{2!} - \frac{4!}{3!} + \frac{4!}{4!} = 12 - 4 + 1 = 9$

Q.11 Answer:

454

Solution:

If all the letters are not in the right envelopes, then at least two letters must be in wrong envelopes.
$\therefore \mathrm{x} = 6! - 1 = 719$
Now y = number of ways so that all the letters are in wrong envelopes =
$6!\left\{1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!}\right\}$
[Derangement formula] $= 360 - 120 + 30 - 6 + 1 = 265$
$\therefore \mathrm{x} - \mathrm{y} = 454$

Q.12 Answer:

119

Solution:

Required number of ways $\sum_{r=2}^{5} {}^5C_{5-r} D(r)$
$= \sum_{r=2}^{5} \frac{5!}{r!(5-r)!} r!\left\{1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \ldots + \frac{(-1)^r}{r!}\right\}$
$= 10 + 20 + (60 - 20 + 5) + (60 - 20 + 5 - 1)$
$= 10 + 20 + 45 + 44 = 119$

Q.13 Answer:

11

Solution:

Since total number of digits $= 6$
So to get a number of a number of the form $\frac{p}{q}$ we will have to choose two numbers out of the 6 numbers.
There is only one way to arrange them since the number x has to satisfy o Hence no. $x = \begin{pmatrix} 6 \\ 2 \end{pmatrix} \times 1 = 15$, but some numbers have to be removed because they are repeated.
They are :

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$\frac{2}{4} = \frac{1}{2}$
$\frac{4}{6} = \frac{2}{3}$
$\frac{2}{6} = \frac{1}{3}$
$\frac{3}{6} = \frac{1}{2}$

Thus the number of removed numbers $= 4$

Therefore, total no. of distinct $x = 15 - 4 = 11$

Hence, option 'D' is correct.

Q.14 Answer:

4

Solution:

$^{20}C_1 = {}^{20}C_{19} = 20$
$^{20}C_3 = {}^{20}C_{17} = 20 \times 19 \times 3$
$^{20}C_5 = {}^{320}C_{15} = 19 \times 17 \times 16 \times 3$
$^{20}C_7 = {}^{20}C_{13} = 19 \times 17 \times 16 \times 15$
$^{20}C_9 = {}^{20}C_{11} = 19 \times 17 \times 13 \times 11 \times 5 \times 8$

Clearly G.C.D. $= 4$

Q.15 Answer:

$^{4n}C_{2n}$

Solution:

Here, $E = (2n+1)(2n+3)(2n+5)\ldots\ldots(4n-3)$
$(4n-1)$
$(2n+1)\cdot(2n+2)\cdot(2n+3)\cdot(2n+4)$
or $E = \frac{\ldots\ldots(4n-1)(4n)}{(2n+2)(2n+4)\ldots(4n)} \cdot \frac{(2n)!}{(2n)!}$
$E = \frac{(4n)!\cdot n!}{(2n)!2^n\cdot(2n)!}$
$\Rightarrow 2^n E = \frac{(4n)!}{(2n)!(2n)!}\cdot n! = {}^{4n}C_{2n}\cdot n!$
$\Rightarrow 2^n E$ **is divisible by** $^{4n}C_{2n}$**.**

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